

Musab Ahmed Khan

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Education

Sabanci University | B.Sc. Computer Science and Engineering | Istanbul, Turkey

Sep 2022 - Jun 2026

CGPA: 3.89/4.0 (High Honors) | 100% Merit Scholarship (Sakip Sabanci)

Key Coursework: Data Science, Spatial Data Science, Machine Learning, Deep Learning, LLMs, Artificial Intelligence, NLP, Network Science, Data Analytics and Optimization, Software Engineering, Automated Debugging, Mobile Application Development, Data Structures and Algorithms, Database Systems, Linear Algebra, Discrete Math, Probability, Statistics

Volunteering: TOM@University, SUBUH Super Active member

Bal Bhawan School | High School | Bhopal, India

Jun 2021

Grade: 90%

Key Coursework: Physics, Chemistry, Maths, Computer Science

Research Experience

Teknofest 2026 - AI in Healthcare: DeepMyelinAI - MS Clinical Decision Support System

Mar 2026 - Present

- Developing an Explainable-AI based Clinical Decision Support System (**CDSS**) for Teknofest 2026 for early diagnosis, disease subtype classification, progression prediction and relapse prediction in Multiple Sclerosis (**MS**) using CSF biomarkers, clinical disability scores, genetic risk profiles, longitudinal symptom histories, and demographic data.
- Designing a multi-stage ML framework targeting the four clinical objectives, employing specialized model types at different stages while ensuring complete transparency for clinical use with **xAI** methods like SHAP, LIME and attention visualization.

Anatomy-Aware Multi-Stage Medical Image Segmentation

Oct 2025 - Feb 2026

- Developed medical image segmentation **CV** models for lung pneumothorax (SIIM-ACR) and brain MRI FLAIR (LGG segmentation), achieving Dice **0.85** and **0.92** respectively. Implemented a lightweight routing classifier to dispatch images to these models automatically with perfect accuracy.
- Implemented UNet++ architecture with EfficientNet (B1 for lung and B3 for brain) encoders using hybrid BCE+Dice loss and **SCSE attention** modules, with class rebalancing, threshold optimization and test-time augmentation (**TTA**) for robust inference and maxpool downsampling to preserve small pathological regions.
- Extended pipeline to include modular post-training reliability flagging mechanism combining MC Dropout uncertainty, domain-shift detection and morphological plausibility checks and compared performance and robustness against heterogeneous mixed model approaches.
- Future plans to add more model types and integrate **xAI** methods (attention maps, saliency) to support clinical decision-making.
- Github:** <https://github.com/orgs/CS415-MultiModal-Med-Image-Segmentation/repositories>

SemEval 2026 Task 5: Ambistory Word Sense Plausibility Prediction

Oct 2025 - Feb 2026

- Built a transformer-based **NLP** model for SemEval 2026 Task 5 to predict human plausibility ratings (1–5) for word senses in ambiguous narrative contexts.
- Used the AmbiStory dataset (2K stories), with future plans including using LLMs to generate augmented datasets in Ambistory's style to reduce overfitting.
- Achieved **0.73** combined score and ranked **14th** on the SemEval leaderboard using fine-tuned **DeBERTa** and novel hybrid CORAL + MSE hybrid loss with dynamic weighting.
- Github:** <https://github.com/NLP-445-SemEval/SU-NLP-29>

Graduation Project: AI Research Gap Finder

Feb 2025 - Jan 2026

- Developed an **AI**-powered end to end human-in-the-loop framework to automate Systematic Mapping Studies (**SMS**) using **LLMs** and Model Context Protocol (**MCP**) agents.
- Made an end to end pipeline of automating months-long SMS process of research paper retrieval, screening with snowballing, topic extraction and classification, and knowledge graph construction for use in research trend and gap analysis, along with smart conference style SMS report generation.

Publications and Preprints

Camera Ready / Accepted

- Khan, M. A. SU NLP 29 at SemEval-2026 Task 5: DynaOrd - Hybrid Dynamic Ordinal Regression with LoRA-Fine-Tuned DeBERTa-v3. SemEval 2026 (co-located with ACL 2026).

Under Review / Submitted

- M. A. Khan and M. S. Salem, "Anatomy-Aware Routing for Robust Medical Image Segmentation: A Failure-Detectable Alternative to Mixed Models," under review at IEEE SIU 2026.

Projects

Spatial Analysis of Political Disorder in Kenya (1997–2024)

Oct 2025 - Jan 2026

- Analyzed 16,225 **ACLED** political violence and protest events (1997–2024) using spatial geometry and OpenStreetMap data, replicating election-cycle patterns.
- Demonstrated an accessibility gradient in political disorder, with riots concentrated near major roads (median ≈ 0.35 km) and battles occurring in remote regions, and quantified systematic coastal vs. inland differences in event frequency and severity.
- **Github:** <https://github.com/Spatial-Data-Science-494/Kenya-Political-Violence-2022>

Github Developer Network Analysis

Jan 2025 - May 2025

- Analyzed 1.9M **GitHub** users and 3.9M follower relationships using GPU-accelerated cuGraph and NetworkX for large-scale developer influence analysis.
- Investigated whether programming language diversity correlates with social network centrality using PageRank, betweenness, and community detection metrics.
- **Github:** <https://github.com/GraphVader/fictional-octo-waddle>

WitchCraft: AI-Assisted CV Manager

May 2025 (8 day project)

- Created a CV-management tool that enables quick creation of professional, tailored, **ATS-friendly** CV with AI assistance if needed.
- **GitHub:** <https://github.com/ProjLab42/WitchCraft>

Understanding a Person's Ethnicity Using a Photograph

Oct 2023 - Jan 2024

- Built an image classification **CNN**-based **CV** model with AI to predict ethnicity from photographs using a dataset of individuals from the GCC, Egypt, and Levant regions.

Work Experience

Eastern AI Maha Company (EMCO), Part-Time Full-Stack Developer | Kuwait City, Kuwait

Nov 2025 - Jan 2026

- Lead further refinement of the MACS 3 auditing system and delivered a more feature-rich, powerful and intuitive solution based on field deployment feedback.
- Collaborated in a 2 person team to enhance all 3 parts of the system to ensure long-term maintainability of the company's nationwide auditing infrastructure. Responsible for tailoring the microservices, database and mobile app to company needs.

Eastern AI Maha Company (EMCO), Software Developer Intern | Kuwait City, Kuwait

Jun 2025 - Aug 2025

- Developed "MACS 3": a 3 component auditing system combining planogram compliance **computer vision model**, mobile task management **app**, and web-based admin **dashboard** to automate FMCG distribution operations of the company across Kuwait.
- Improved compliance accuracy by **67%** within 10 days, with **80%** of non-compliance flags now originating from the mobile app compared to manual reporting.
- Responsible for developing Backend Kubernetes microservices task, photos, override requests and reports along with PostgreSQL database structure and automated database management system. Also responsible for developing Flutter Mobile App.

Skills

Natural Languages: English – Professional Proficiency | **Hindi** – Native Proficiency | **Urdu** – Native Proficiency

Computing Languages: C++, Python, R, Java, Prolog, Verilog, SQL, TypeScript, JavaScript, HTML & CSS

Technical: MySQL, PostgreSQL, MongoDB, Firebase, CUDA, Numba, Pandas, PyTorch, Scikit-learn, Numpy, NetworkX, cuGraph, UNet++, Node.js, Express.js, React.js, React Native, Git, Github, Jest, Kubernetes, Docker, Skaffold, Android Studio, Ollama, Postman, Scrum, Jira, LaTeX

Soft: Hardworking, Creativity, Critical thinking, Time management, Teamwork, Communication, Problem solving, Empathy

Certificates

NVIDIA - Fundamentals of Accelerated Computing with CUDA Python

May 2025

NVIDIA - Applications of AI for Predictive Maintenance

Feb 2025

NVIDIA - Generative AI with Diffusion Models

Mar 2024

NVIDIA - Building Transformer-Based Natural Language Processing Applications

Mar 2024

NVIDIA - Fundamentals of Accelerated Computing with CUDA C/C++

Mar 2024

IBM - Data Analysis with Python

Oct 2022

MITx X Series Program - Computational Thinking using Python

Dec 2021

HarvardX CS50AI - CS50's Introduction to Artificial Intelligence with Python

Aug 2021